

SARGATSKOE, Russian Federation

WMO: 285980

Lat: 55.600N

Lon: 73.483E

Elev: 81

StdP: 100.36

Time Zone: 6.00 (E06)

Period: 90-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.5	(c) -32.1	(d) -38.1	(e) 0.1	(f) -35.5	(g) -34.7	(h) 0.1	(i) -31.9	(j) 10.4	(k) -4.2	(l) 9.4	(m) -4.8	(n) 1.5	(o) 210	(p) 0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 30.2	(d) 19.1	(e) 28.4	(f) 18.6	(g) 26.6	(h) 17.8	(i) 20.8	(j) 27.5	(k) 19.8	(l) 25.9	(m) 18.9	(n) 24.7	(o) 3.8	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	23.1	17.6	12.7	22.3	16.7	12.0	21.4	60.6	27.6	57.1	26.0	54.0	24.7	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.0	(b) 8.8	(c) 7.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-16.4	-15.9	-6.9	3.1	12.0	17.1	19.1	16.4	8.8	3.1	-8.2	-15.6
	DBStd	14.33	8.04	8.13	6.35	6.20	5.68	4.75	3.56	4.11	4.14	5.08	7.91	7.11
	HDD10.0	3964	818	726	522	219	45	5	0	3	70	217	547	792
	HDD18.3	6287	1077	959	781	458	207	78	34	87	286	472	797	1051
	CDD10.0	857	0	0	0	11	107	218	282	200	34	4	0	0
	CDD18.3	137	0	0	0	0	11	42	58	26	1	0	0	0
	CDH23.3	1389	0	0	0	2	158	453	506	253	17	0	0	0
	CDH26.7	377	0	0	0	1	38	129	138	68	2	0	0	0
(14) Wind	WSAvg	3.3	3.5	3.4	3.6	3.7	3.6	3.0	2.7	2.6	3.1	3.7	3.5	3.5
(15) Precipitation	PrecAvg	395	18	15	17	22	30	53	65	56	31	31	30	25
	PrecMax	550	42	30	45	51	72	96	133	135	83	76	56	47
	PrecMin	259	3	2	2	4	11	17	15	17	9	6	9	11
	PrecStd	72	10	6	11	13	15	22	33	26	16	16	13	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	1.4	5.7	22.2	30.2	31.6	32.9	31.6	26.3	18.1	6.3	0.8
		MCWB	0.2	0.0	2.6	13.1	17.4	18.9	21.0	19.7	15.1	9.9	4.9	0.5
	2%	DB	-1.0	-0.2	3.8	18.4	27.1	29.9	30.0	28.4	21.3	15.1	4.6	-0.9
		MCWB	-1.8	-1.2	1.8	11.2	15.9	18.8	19.5	18.7	13.5	9.3	3.3	-1.2
	5%	DB	-2.6	-1.3	2.4	15.6	24.9	28.0	28.1	25.7	18.3	12.4	2.5	-3.4
		MCWB	-3.2	-2.0	1.1	9.7	14.7	18.4	19.2	17.7	12.2	8.1	1.4	-4.0
	10%	DB	-4.8	-3.2	1.4	12.8	22.4	26.0	26.2	23.8	15.5	10.4	1.1	-5.6
		MCWB	-5.3	-3.7	0.3	8.2	14.1	17.6	18.7	16.9	11.3	7.0	0.3	-6.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.2	3.3	14.3	18.7	22.0	22.0	21.1	16.0	12.0	5.5	0.4
		MCDB	0.8	1.0	4.9	19.9	28.5	29.3	28.4	28.0	23.7	16.1	6.2	0.6
	2%	WB	-1.8	-1.0	2.1	12.2	16.6	20.3	21.2	19.6	14.2	10.0	3.3	-1.2
		MCDB	-1.3	-0.3	3.3	17.4	24.7	27.0	27.9	26.5	19.3	13.7	4.3	-0.8
	5%	WB	-3.3	-2.1	1.3	10.2	15.5	19.1	20.2	18.6	12.9	8.8	1.5	-4.1
		MCDB	-2.6	-1.4	2.3	15.0	23.4	25.7	26.1	24.2	16.8	11.8	2.3	-3.6
	10%	WB	-5.4	-3.7	0.5	8.3	14.3	18.1	19.3	17.6	11.6	7.5	0.3	-6.2
		MCDB	-4.8	-3.1	1.4	12.6	21.3	24.8	24.7	22.2	15.4	10.1	1.0	-5.6
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.1	9.7	9.9	12.2	10.9	9.9	10.7	9.2	7.3	7.1	8.3
		MCDBR	7.7	7.3	7.9	14.6	16.2	14.3	12.8	14.0	14.1	10.9	5.8	8.4
		MCWBR	7.5	6.9	6.5	8.8	7.1	6.0	4.8	5.8	7.3	6.8	5.4	8.2
	5% WB	MCDBR	7.9	7.2	6.9	13.9	14.8	12.9	11.0	11.6	12.0	10.1	5.8	7.9
		MCWBR	7.6	6.9	6.1	8.9	7.2	6.3	4.9	5.7	7.1	6.9	5.5	7.8
(40) Clear-Sky Solar Irradiance	taub		0.282	0.316	0.375	0.411	0.403	0.411	0.432	0.432	0.371	0.355	0.321	0.287
	taud		2.038	2.012	1.922	2.095	2.226	2.282	2.220	2.187	2.322	2.220	2.085	1.991
	Ebn at Noon		596	720	759	801	837	835	805	775	780	678	536	474
	Edh at Noon		66	100	141	138	129	124	129	125	95	82	64	55
(44) All-Sky Solar Radiation	RadAvg		0.90	1.97	3.54	4.62	5.49	5.90	5.46	4.34	2.99	1.54	0.76	0.58
	RadStd		0.07	0.10	0.25	0.35	0.46	0.40	0.39	0.37	0.27	0.17	0.12	0.06

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page